SAFETY DATA SHEET

According to Regulation (EC) No. 1907/2006 (REACH) Article 31, Annex II as amended 2015/830.

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Product name: Ethyl Acetate

Product No. 5277, 9260, 8038, 8037, 6780, H078

Synonyms, Trade Names: Acetic acid ethyl ester; Ethyl ethanoate; Acetoxyethane; Acetic ether

Additional identification

Chemical name:	Ethyl acetate
Chemical formula:	C4H8O2
INDEX No.	607-022-00-5
CAS-No.	141-78-6
EC No.	205-500-4
REACH Registration No.	01-2119475103-46-XXXX

1.2 Relevant identified uses of the substance or mixture and uses advised against

Identified uses: For Laboratory, Research or Manufacturing Use.
Uses advised against: Not determined.

1.3 Details of the supplier of the safety data sheet

Avantor Performance Materials Poland S.A.
Sowinskiego 11str., 44-101 Gliwice,
Poland

Telephone: 48 32 239-20-00
Fax: 48 32 239-23-70

Contact person: Product Information Compliance
E-mail: export@avantormaterials.com

E-mail address of person responsible for this SDS: SDS@avantormaterials.com

1.4 Emergency telephone number: CHEMTREC: (44)-870-8200418

SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

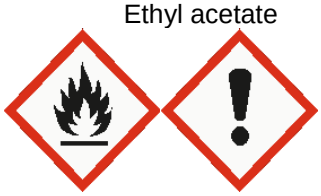
The product has been classified according to the legislation in force.

Classification according to Regulation (EC) No 1272/2008 as amended.

Physical Hazards		
Flammable liquids	Category 2	H225: Highly flammable liquid and vapour.
Health Hazards		
Serious eye irritation	Category 2	H319: Causes serious eye irritation.
Specific Target Organ Toxicity - Single Exposure	Category 3	H336: May cause drowsiness or dizziness.

2.2 Label Elements

Contains: Ethyl acetate



Signal Word: Danger

Hazard Statement(s):
H225: Highly flammable liquid and vapour.
H319: Causes serious eye irritation.
H336: May cause drowsiness or dizziness.
EUH066: Repeated exposure may cause skin dryness or cracking.

Target organs:
Ethyl acetate narcotic effects

Precautionary Statements

Prevention:
P210: Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
P241: Use explosion-proof [electrical/ventilating/lighting] equipment.

Response:
P303+P361+P353: IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water [or shower].
P305+P351+P338: IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

Storage:
P405: Store locked up.

2.3 Other hazards No data available.

SECTION 3: Composition/information on ingredients

3.1 Substances

Chemical name	Concentration	CAS-No.	EC No.	REACH Registration No.	M-Factor:	Notes
Ethyl acetate	50 - <100%	141-78-6	205-500-4	01-2119475103-46-XXXX	No data available.	#

* All concentrations are percent by weight unless ingredient is a gas. Gas concentrations are in percent by volume.
This substance has workplace exposure limit(s).

SECTION 4: First Aid Measures

General: Get medical advice/attention if you feel unwell. Show this safety data sheet to the doctor in attendance.

4.1 Description of first aid measures

Inhalation: Move to fresh air. Get medical attention if symptoms persist.

Skin Contact: Wash skin thoroughly with soap and water. Get medical attention if symptoms occur. Wash contaminated clothing before re-use.

Eye contact:	Immediately flush with plenty of water for at least 15 minutes. If easy to do, remove contact lenses. Get medical attention.
Ingestion:	Call a doctor or poison control centre immediately. Do NOT induce vomiting. If vomiting occurs, keep head low so that stomach content doesn't get into the lungs.
4.2 Most important symptoms and effects, both acute and delayed:	Irritating to eyes, respiratory system and skin.

4.3 Indication of any immediate medical attention and special treatment needed

Hazards:	No data available.
Treatment:	Treat symptomatically. Symptoms may be delayed.

SECTION 5: Firefighting Measures

General Fire Hazards:	Flammable liquid and vapour. Can be ignited easily and burns vigorously.
5.1 Extinguishing media **Suitable extinguishing media:**	Water spray, foam, dry powder or carbon dioxide.
Unsuitable extinguishing media:	Avoid water in straight hose stream; will scatter and spread fire.
5.2 Special hazards arising from the substance or mixture:	Can be ignited easily and burns vigorously. Vapours may cause a flash fire or ignite explosively. Vapours may travel considerable distance to a source of ignition and flash back. Prevent buildup of vapours or gases to explosive concentrations.
5.3 Advice for firefighters **Special fire fighting procedures:**	Move containers from fire area if you can do so without risk. Use water spray to keep fire-exposed containers cool. Cool containers exposed to flames with water until well after the fire is out. Water may be ineffective in fighting the fire. Fight fire from a protected location. Flammable liquid and vapour. Can be ignited easily and burns vigorously.
Special protective equipment for firefighters:	Firefighters must use standard protective equipment including flame retardant coat, helmet with face shield, gloves, rubber boots, and in enclosed spaces, SCBA.

SECTION 6: Accidental Release Measures

6.1 Personal precautions, protective equipment and emergency procedures:	Keep unauthorised personnel away. Keep upwind. Ventilate closed spaces before entering them. Use personal protective equipment. Do not touch damaged containers or spilled material unless wearing appropriate protective clothing. See Section 8 of the SDS for Personal Protective Equipment.
6.2 Environmental Precautions:	Do not contaminate water sources or sewer. Prevent further leakage or spillage if safe to do so. Avoid discharge into drains, water courses or on to the ground.
6.3 Methods and material for containment and cleaning up:	Absorb spill with vermiculite or other inert material, then place in a container for chemical waste. Dyke far ahead of larger spill for later recovery and disposal. In case of leakage, eliminate all ignition sources. Use non-sparking tools. All equipment used when handling the product must be grounded.

6.4 Reference to other sections: No data available.

SECTION 7: Handling and Storage:

7.1 Precautions for safe handling: DO NOT handle, store or open near an open flame, sources of heat or sources of ignition. Protect material from direct sunlight. Take precautionary measures against static discharges. Use personal protective equipment as required. Avoid breathing mists or vapours. Avoid contact with eyes, skin, and clothing. Do not taste or swallow. Use only with adequate ventilation. Wash hands thoroughly after handling. See Section 8 of the SDS for Personal Protective Equipment.

7.2 Conditions for safe storage, including any incompatibilities: Keep away from food, drink and animal feeding stuffs. Keep container tightly closed in a cool, well-ventilated place. Ground container and transfer equipment to eliminate static electric sparks. Comply with all national, state, and local codes pertaining to the storage, handling, dispensing, and disposal of flammable liquids.

7.3 Specific end use(s): No data available.

SECTION 8: Exposure Controls/Personal Protection

8.1 Control Parameters

Occupational Exposure Limits

Chemical name	Type	Exposure Limit Values	Source
Ethyl acetate	TWA	200 ppm	UK. EH40 Workplace Exposure Limits (WELs) (2007)
	STEL	400 ppm	UK. EH40 Workplace Exposure Limits (WELs) (2007)
	TWA	200 ppm 734 mg/m3	EU. Indicative Exposure Limit Values in Directives 91/322/EEC, 2000/39/EC, 2006/15/EC, 2009/161/EU (02 2017)
	STEL	400 ppm 1.468 mg/m3	EU. Indicative Exposure Limit Values in Directives 91/322/EEC, 2000/39/EC, 2006/15/EC, 2009/161/EU (02 2017)
	STEL	400 ppm 1.468 mg/m3	EU. Scientific Committee on Occupational Exposure Limit Values (SCOELs), European Commission - SCOEL (2014)
	TWA	200 ppm 734 mg/m3	EU. Scientific Committee on Occupational Exposure Limit Values (SCOELs), European Commission - SCOEL (2014)

DNEL-Values

Critical component	Type	Route of Exposure	Health Warnings	Remarks
Ethyl acetate	Workers	Inhalation	Systemic, short-term; 1468 mg/m3	Respiratory tract irritation.

	General population		Systemic, short-term; 734 mg/m3	Respiratory tract irritation.
			Local, long-term; 367 mg/m3	Respiratory tract irritation.
	Workers		Systemic, long-term; 734 mg/m3	Respiratory tract irritation.
	General population		Local, short-term; 734 mg/m3	Respiratory tract irritation.
	Workers		Local, short-term; 1468 mg/m3	Respiratory tract irritation.
			Local, long-term; 734 mg/m3	Respiratory tract irritation.
		Dermal	Systemic, long-term; 63 mg/kg bw/day	Respiratory tract irritation.
	General population	Inhalation	Systemic, long-term; 367 mg/m3	Respiratory tract irritation.
		Dermal	Systemic, long-term; 37 mg/kg bw/day	Respiratory tract irritation.
		Eyes	Local effect;	No data available
		Oral	Systemic, long-term; 4,5 mg/kg bw/day	Respiratory tract irritation.
	Workers	Eyes	Local effect;	

PNEC-Values

Critical component	Environmental compartment	PNEC-Values
Ethyl acetate	Sewage treatment plant	650 mg/l
	Sediment (freshwater)	1,15 mg/kg
	Sediment (marine water)	0,115 mg/kg
	Predator	0,2 g/kg
	Aquatic (freshwater)	0,24 mg/l
	Aquatic (marine water)	0,024 mg/l
	Soil	0,148 mg/kg

8.2 Exposure controls

Appropriate Engineering Controls: No data available.

Individual protection measures, such as personal protective equipment

General information: Good general ventilation (typically 10 air changes per hour) should be used. Ventilation rates should be matched to conditions. If applicable, use process enclosures, local exhaust ventilation, or other engineering controls to maintain airborne levels below recommended exposure limits. If exposure limits have not been established, maintain airborne levels to an acceptable level. An eye wash and safety shower must be available in the immediate work area. Use explosion-proof ventilation equipment.

Eye/face protection: Wear safety glasses with side shields (or goggles) and a face shield.

Skin protection

Hand Protection: Material: Chemical resistant gloves

Other: Wear suitable protective clothing.

Respiratory Protection: If engineering controls do not maintain airborne concentrations below recommended exposure limits (where applicable) or to an acceptable level (in countries where exposure limits have not been established), an approved respirator must be worn. Respirator type: Chemical respirator with organic vapour cartridge and full facepiece.

Hygiene measures: Provide eyewash station and safety shower. Always observe good personal hygiene measures, such as washing after handling the material and before eating, drinking and/or smoking. Routinely wash work clothing to remove contaminants. Discard contaminated footwear that cannot be cleaned.

Environmental Controls: No data available.

SECTION 9: Physical And Chemical Properties

9.1 Information on basic physical and chemical properties

Appearance

Physical state:	Liquid
Form:	Liquid
Colour:	Colorless
Odour:	Fruity
Odour Threshold:	No data available.
pH:	No data available.
Freezing point:	-83 °C
Boiling Point:	77 °C
Flash Point:	-4 °C (Closed Cup)
Evaporation Rate:	4,1 6 (butyl acetate=1)
Flammability (solid, gas):	Class IB Flammable Liquid
Flammability limit - upper (%)	11,5 %(V)
Flammability limit - lower (%)	2 %(V)
Vapour pressure:	12,43 kPa (25 °C) 98 hPa (20 °C) 379 hPa (50 °C)
Vapour density (air=1):	3,04 Air=1
Density:	0,90 g/ml (20 °C)
Relative density:	0,902 (20 °C)
Solubility(ies)	
Solubility in Water:	60 - 80 g/l (25 °C)
Solubility (other):	No data available.
Partition coefficient (n-octanol/water):	0,73
Autoignition Temperature:	427 °C
Decomposition Temperature:	No data available.
Viscosity:	0,417 - 0,648 mm2/s (25 °C)
Explosive properties:	No data available.
Oxidising Properties:	No data available.

9.2 Other information

Molecular weight:	88,11 g/mol (C4H8O2)
VOC content:	EC Directive 2004/42: 900 g/l ~100 % (calculated)
Minimum ignition energy:	0,46 mJ
Minimum ignition temperature:	485 °C
Metal corrosion:	

SECTION 10: Stability and Reactivity

10.1 Reactivity: No dangerous reaction known under conditions of normal use.

10.2 Chemical Stability: Material is stable under normal conditions.

10.3 Possibility of Hazardous Reactions: Hazardous polymerization does not occur.

10.4 Conditions to Avoid: Heat, sparks, flames.

10.5 Incompatible Materials: Strong oxidising agents. Strong alkalis. Strong acids. Nitrates. May attack some plastics, rubber and coatings.

10.6 Hazardous Decomposition Products: Thermal decomposition may release oxides of carbon.

SECTION 11: Toxicological Information

Information on likely routes of exposure

Inhalation: None known or expected under normal use.

Skin Contact: None known or expected under normal use.

Eye contact: Causes serious eye irritation.

Ingestion: None known or expected under normal use.

11.1 Information on toxicological effects

Acute toxicity

Oral
Product: LD 50 (Rat): 5,6 g/kg

Dermal
Product: LD 50 (Rabbit) > 20.000 mg/kg

Inhalation
Product: LD 50 (Rat, 4 h): 4000 ppm

Repeated dose toxicity
Product: No data available.

Skin Corrosion/Irritation:
Product: Causes mild skin irritation.

Serious Eye Damage/Eye Irritation:
Product: Causes serious eye irritation.

Respiratory or Skin Sensitisation:
Product: Not a skin nor a respiratory sensitizer.

Germ Cell Mutagenicity

In vitro
Product: No mutagenic components identified

In vivo
Product: No mutagenic components identified

Carcinogenicity
Product: This substance has no evidence of carcinogenic properties.

Reproductive toxicity
Product: No components toxic to reproduction

Specific Target Organ Toxicity - Single Exposure
Product: Narcotic effect. Respiratory tract irritation.

Specific Target Organ Toxicity - Repeated Exposure
Product: None known.

Target organs:
Ethyl acetate narcotic effects

Aspiration Hazard
Product: May be harmful if swallowed and enters airways.

Other Adverse Effects: None known.

SECTION 12: Ecological Information

12.1 Toxicity

Acute toxicity

Fish
Product: No data available.

Specified substance(s)
Ethyl acetate
LC 50 (Pimephales promelas, 96 h): 230 mg/l
LC 50 (Oncorhynchus mykiss, 96 h): 230 - 500 mg/l
EC 50 (Pimephales promelas, 96 h): 220 mg/l
LC 50 (Heteropneustes fossilis, 96 h): 212,5 mg/l
LOAEL (Pimephales promelas, 96 h): 19,4 mg/l

Aquatic Invertebrates
Product: No data available.

Specified substance(s)
Ethyl acetate
EC 50 (Brine shrimp (Artemia salina), 24 h): 306,9 - 389,9 mg/l
LC 50 (Water flea (Daphnia magna), 48 h): 560 - 819 mg/l
LC 50 (Hydra (Hydra oligactis), 48 h): 1.350 mg/l
LC 50 (Water flea (Daphnia cucullata), 48 h): 154 - 175 mg/l
EC 50 (Water flea (Daphnia magna), 48 h): 610 - 799 mg/l

Chronic toxicity

Fish
Product: No data available.

Specified substance(s)
Ethyl acetate
NOAEL (Pimephales promelas, 32 d): < 9,65 mg/l

Aquatic Invertebrates
Product: No data available.

Specified substance(s)
Ethyl acetate
NOAEL (Daphnia magna, 21 d): 2,4 mg/l (semi-static)

Toxicity to aquatic plants
Product: No data available.

Specified substance(s)
Ethyl acetate — No data available.

12.2 Persistence and Degradability

Biodegradation
Product: Expected to be readily biodegradable.

Specified substance(s)
Ethyl acetate — No data available.

BOD/COD Ratio
Product No data available.

Specified substance(s)
Ethyl acetate — No data available.

12.3 Bioaccumulative Potential
Product: No data available on bioaccumulation.

Specified substance(s)
Ethyl acetate — Green algae (Chlorella fusca vacuolata), Bioconcentration Factor (BCF): 13.500 (Static)
Leuciscus idus, Bioconcentration Factor (BCF): 30 Aquatic sediment
Rat, Bioconcentration Factor (BCF): 8,6
Bioconcentration Factor (BCF): 3.300 Aquatic sediment
Chlorella fusca var. vacuolata, Bioconcentration Factor (BCF): 13.500 Aquatic sediment

12.4 Mobility in Soil: The product is water soluble and may spread in water systems.
Known or predicted distribution to environmental compartments
Ethyl acetate — No data available.

12.5 Results of PBT and vPvB assessment: Not available.
Ethyl acetate — No data available.

12.6 Other Adverse Effects: The product components are not classified as environmentally hazardous. However, this does not exclude the possibility that large or frequent spills can have a harmful or damaging effect on the environment.

12.7 Additional Information: No data available.

SECTION 13: Disposal Considerations

13.1 Waste treatment methods

General information: No data available.

Disposal methods: Discharge, treatment, or disposal may be subject to national, state, or local laws.

SECTION 14: Transport Information

ADR

14.1 UN Number:	UN 1173
14.2 UN Proper Shipping Name:	ETHYL ACETATE
14.3 Transport Hazard Class(es)	
Class:	3
Label(s):	3
Hazard No. (ADR):	33
Tunnel restriction code:	(D/E)
14.4 Packing Group:	II
14.5 Environmental Hazards:	No
14.6 Special precautions for user:	Not determined.

RID

14.1 UN Number:	UN 1173
14.2 UN Proper Shipping Name	ETHYL ACETATE
14.3 Transport Hazard Class(es)	
Class:	3
Label(s):	3
14.4 Packing Group:	II
14.5 Environmental Hazards:	No
14.6 Special precautions for user:	Not determined.

IMDG

14.1 UN Number:	UN 1173
14.2 UN Proper Shipping Name:	ETHYL ACETATE
14.3 Transport Hazard Class(es)	
Class:	3
Label(s):	3
EmS No.:	F-E, S-D
14.4 Packing Group:	II
14.5 Environmental Hazards:	No
14.6 Special precautions for user:	Not determined.

IATA

14.1 UN Number:	UN 1173
14.2 Proper Shipping Name:	Ethyl acetate
14.3 Transport Hazard Class(es):	
Class:	3
Label(s):	3
14.4 Packing Group:	II
14.5 Environmental Hazards:	No
14.6 Special precautions for user:	Not determined.

14.7 Transport in bulk according to Annex II of MARPOL and the IBC Code: Not applicable

SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture:

EU Regulations

Regulation (EC) No. 2037/2000 Substances that deplete the ozone layer: none

Regulation (EC) No. 850/2004 on persistent organic pollutants: none

Regulation (EC) No. 649/2012 Import and export of dangerous chemicals: none

Regulation (EC) No. 1907/2006, REACH Annex XIV Substances subject to authorisation, as amended: none

Regulation (EC) No. 1907/2006 Annex XVII Substances subject to restriction on marketing and use:

Chemical name	CAS-No.	Concentration
Ethyl acetate	141-78-6	100%

Directive 2004/37/EC on the protection of workers from the risks related to exposure to carcinogens and mutagens at work.: none

Directive 92/85/EEC: on the safety and health of pregnant workers and workers who have recently given birth or are breast feeding.: none

Directive 2012/18/EU (Seveso III): on the control of major accident hazards involving dangerous substances:

Chemical name	CAS-No.	Concentration
Ethyl acetate	141-78-6	100%

EU. Regulation No. 166/2006 PRTR (Pollutant Release and Transfer Registry), Annex II: Pollutants: none

Directive 98/24/EC on the protection of workers from the risks related to chemical agents at work:

Chemical name	CAS-No.	Concentration
Ethyl acetate	141-78-6	100%

Listed on the Candidate List of substances of very high concern (SVHC) Regulation (EC) No. 1907/2006, REACH Article 59(1). Candidate List

Contains: none

15.2 Chemical safety assessment: No Chemical Safety Assessment has been carried out.

Inventory status:

AICS:	On or in compliance with the inventory
DSL:	On or in compliance with the inventory
IECSC:	On or in compliance with the inventory
ENCS (JP):	On or in compliance with the inventory
ISHL (JP):	On or in compliance with the inventory
KECI (KR):	On or in compliance with the inventory
INSQ:	On or in compliance with the inventory
NZIOC:	On or in compliance with the inventory
PICCS (PH):	On or in compliance with the inventory
TCSI:	On or in compliance with the inventory
TSCA:	On or in compliance with the inventory
EU INV:	On or in compliance with the inventory

SECTION 16: Other Information

Revision Information: Not relevant.

References

PBT	PBT: persistent, bioaccumulative and toxic substance.
vPvB	vPvB: very persistent and very bioaccumulative substance.

Key literature references and sources for data: Sources of information used in preparing this SDS included one or more of the following: results from in house or supplier toxicology studies, information from the Toxicology Data Network (TOXNET), European Chemical Agency (ECHA) substance dossiers, IARC Monographs, US National Toxicology Program data, the Agency for Toxic Substances and Disease Registry, other manufacturer's SDSs and other sources, as appropriate.

Wording of the H-statements in sections 2 and 3

H225	Highly flammable liquid and vapour.
H319	Causes serious eye irritation.
H336	May cause drowsiness or dizziness.

Training information: No data available.

Issue Date: 09.06.2021

SDS No.:

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